Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **React** and Next.js web apps with **TypeScript**, RESTful services, and PostgreSQL and SQL Server schemas, plus automated testing, CI/CD on AWS, and Agile delivery across AI, education, and search products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized **PostgreSQL** schemas with indexing and **query tuning** for 200+ users and 10K+ records, running SQL transformations across PostgreSQL and SQL Server that cut latency 35%.
- Built RESTful FastAPI services and integration pipelines with validation and CI/CD-backed releases at 99%+ reliability, and supported production by isolating failing jobs to triple throughput across 1K+ async jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **React** education product with REST APIs returning JSON and MySQL on AWS, deployed through GitHub Actions CI/CD and serving 150+ instructors and 10K+ daily requests.
- Integrated GPT-4o, Claude, and Qwen-VL into rubric generation and grading workflows with OCR fallbacks, measuring 90%+ grading consistency across 10K+ submissions while cutting instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ automated **unit and integration test** suites with PyTest and Mockito, run nightly in Jenkins across Windows, Linux, and macOS, owning quality through every release with coverage reports.
- Broke stakeholder requirements into sprint work with Microsoft and GitHub teams in **Agile** iterations, and optimized web services with pagination and caching to improve response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **TypeScript** React platform with RESTful **microservices** for customer records, billing, and leads on Azure SQL, syncing 2M+ records across multiple locations with real-time WebSocket updates for every site.
- Ran unit, integration, and **end-to-end** tests with Jest and Selenium in Jenkins CI/CD, provisioned infrastructure with ARM templates, and deployed Docker images to Azure, cutting deployment time 50% across releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a React and TypeScript dashboard with Flask REST APIs for bookings and pricing, keeping responses under 200ms across 50K+ records with Firebase real-time updates on mobile and web clients.
- Integrated third-party payment services through idempotent webhooks for refunds and disputes, cutting manual handling about 70%, and presented the **architecture** to the client before implementation after mapping 12+ integration points.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, HTML, CSS
Web: React, Next.js, REST APIs, JSON, HTTP, Microservices, Node.js, FastAPI, Flask
Data & Search: PostgreSQL, SQL Server, MySQL, Query Optimization, Schema Design, Semantic Search, FAISS
Testing & DevOps: Jest, Selenium, PyTest, Unit Testing, Integration Testing, CI/CD, Jenkins, GitHub Actions, ARM Templates
Cloud & Practices: AWS, Azure, Docker, Git, Agile, Code Reviews, LLM APIs, Linux

Projects

Research Article Discovery Search | *Next.js, TypeScript, PostgreSQL, Elasticsearch, Jest*

- Designed a **Next.js** search interface for research articles with subject and date facets, full-text relevance ranking, and saved searches, built test-first with Jest and Playwright **component tests**.

Citation Export Service | *C#, .NET, SQL Server, REST APIs, xUnit*

- Designed a versioned **.NET** REST service that formats article metadata into APA, MLA, and Chicago citations, with **contract tests** so client apps stay compatible across API versions.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, REST APIs*

- Built a **document discovery** platform with semantic search, Q&A, and summarization over 50K+ documents, using incremental indexing to cut retrieval latency 40% at 95% query accuracy for 200+ users.